Mitoprep Answer Key 4

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Q1) **Explanation:**

Hint: $\vec{\tau} = \vec{r} \times \vec{F}$

Explanation: Given that both siblings push with the same magnitude of force and perpendicular to the door, the torque exerted by each sibling will depend solely on the distance from the hinge at which they apply their force. Since, the brother pushes farther from the hinge than the sister, he will exert more torque.
Hence, option (2) is the correct answer.

Q2) **Explanation:**

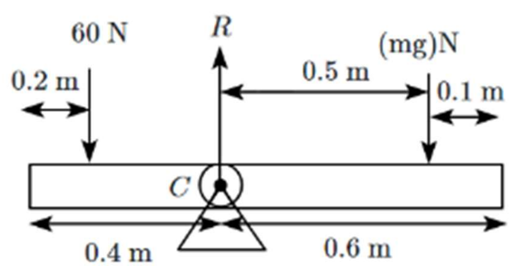
Hint: $\vec{\tau} = \vec{r} \times \vec{F}$

Step: Find the torque of the force $\vec{C}$ about the pivot.
The perpendicular distance of the force $\vec{C}$ from the pivot is zero, so the torque of the force $\vec{C}$ about the pivot is $\tau = r \times F = 0 \times F = 0$
Therefore, the torque of the force $\vec{C}$ about the pivot is zero.
Hence, option **(2)** is the correct answer.

Q3) **Explanation:**

Hint: $F_1 R_1 = F_2 R_2$

Step: Find the required mass to balance the torque.



According to the principle of moments;
$60 \times 0.2 = m \times 10 \times 0.5$
$\Rightarrow 12 = 5m$
$\Rightarrow m = 2.4$ kg
Hence, option (1) is the correct answer.

Q4) **Explanation:**

Hint: $\tau = I\alpha$

Step: Find the angular acceleration of the disc.

$$\tau = I\alpha$$

$$R \times F = \frac{MR^2}{2}(\alpha)$$

$$\Rightarrow \alpha = \frac{2F}{MR}$$

Hence, option (3) is the correct answer.

Q5) **Explanation:**

Hint: $\vec{\tau} = \frac{d\vec{L}}{dt}$

Explanation: A central force field means that the force on each particle is directed towards or away from a common centre (usually taken as the origin) and depends only on the distance from this centre. The mathematical form of central force is given by; $\vec{f} = f(r)\hat{r}$ where, r is the distance from the centre.

The torque acting on the system of particles is given by; $\vec{\tau} = \vec{r} \times \vec{F}$

As both $\vec{r}$ and $\vec{F}$ are parallel (or anti-parallel), leading to $\vec{r} \times \vec{F} = 0$

So the torque acting on the system of particles is zero.

The angular momentum of the system of particles is constant because the external torque acting on the system of particles is zero.

Therefore, Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.

Hence, option (1) is the correct answer.

Q6) **Explanation:**

Hint: There is no external force acting along the horizontal direction.

Step: Find the velocity of the wedge.

Since the system is initially at rest and no external forces act on it horizontally, the total momentum before and after the block reaches the bottom must be conserved. This means the sum of the horizontal component of the momenta of the block and the wedge at any instant must be the same.

$$0 = mu + mv$$

$$\Rightarrow u = -v$$

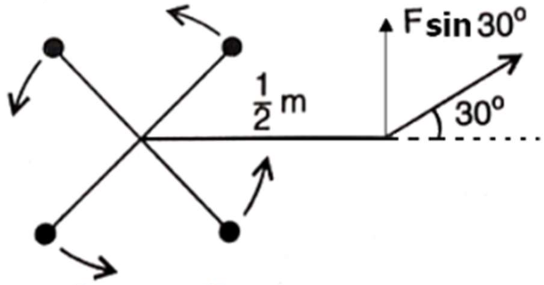
Therefore, the velocity of the wedge when the block reaches the bottom is $-v$, which means the wedge has a horizontal velocity equal to the block but in the opposite direction.

Hence, option (1) is the correct answer.

Q7)

Hint: $\vec{\tau} = I\vec{\alpha}$

Step 1: Find the moment of inertia about center.



$$I = 4\left[(2)\left(\frac{1}{4}\right)^2\right]$$
$$= \frac{1}{2}$$

Step 2: Find the torque.

$\tau = I\alpha$

$24 \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{2}\alpha$

$\alpha = 12\ rad/s^2$

Q8) **Explanation:**

Hint: $L = mvr_{\perp}$

Explanation: If the particle is moving directly away or towards the fixed point (along the line), the angle between r and $p(p = mv)$ is either 0° or 180°, makes the cross product $\vec{r} \times \vec{p}$ equal to zero.

Therefore, the angular momentum L about that point remains zero at all times, because the cross product $r \times p$ zero.

Hence, option **(4)** is the correct answer.

Q9) **Explanation:**

Hint: $F_e = ma_{C.O.M}$

Explanation: The total mass of a system of particles times the acceleration of its centre of mass is the vector sum of all the forces acting on the system of particles. Among these forces on each particle, there will be external forces exerted by bodies outside the system and also internal forces exerted by the particles on one another. We know from Newton's third law that these internal forces occur in equal and opposite pairs, and their contribution is zero in the sum of forces. Only the external forces contribute to the acceleration. The centre of mass of a system of particles moves as if all the mass of the system was concentrated at the centre of mass and all the external forces were applied at that point.

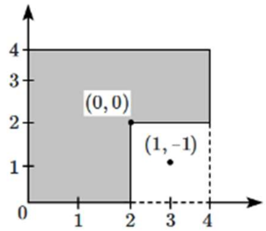
Therefore, both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.

Hence, option **(1)** is the correct answer.

Q10) **Explanation:**

Hint: $X_{\text{com}} = \frac{\sum_{i=1}^{N} m_i x_i}{M}$

Step: Find the shift in the centre-of-mass.



Let the mass of the (2×2) square sheet be m, then the mass of the (4×4) square sheet will be $4m$. Let us consider the centre-of-mass of (4×4) square sheet as the origin, then the coordinates of the (2×2) square sheet will be $(1, -1)$. Now the shift in centre-of-mass is the same as the distance from the origin to the centre-of-mass of the remaining portion.
The centre-of-mass of the remaining portion is given by:

$\vec{r}_{\text{com}} = \frac{4m(0,0) - m(1,-1)}{4m - m}$

$\Rightarrow \vec{r}_{\text{com}} = \left(\frac{-1}{3}, \frac{1}{3}\right)$

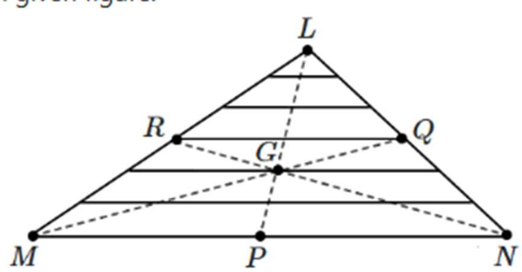
$\Rightarrow |\vec{r}_{\text{com}}| = \sqrt{\left(-\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2} = \frac{\sqrt{2}}{3}$ m

Therefore, the shift in the centre-of-mass is $\frac{\sqrt{2}}{3}$ m.
Hence, option **(1)** is the correct answer.

Q11) **Explanation:**

Hint: Recall the formula for the centre of mass of a triangular lamina.

Step: Find the centre of mass of a thin, uniform lamina.
The lamina ($\triangle LMN$) may be subdivided into narrow strips each parallel to the base (MN) as shown in given figure.

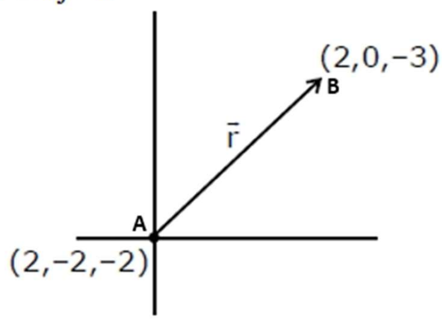


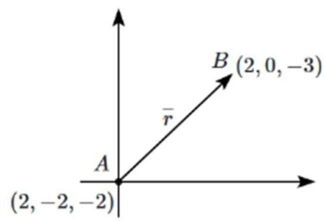
By symmetry each strip has its centre of mass at its midpoint. If we join the midpoint of all the strips we get the median LP. The centre of mass of the triangle as a whole therefore, has to lie on the median LP.
Similarly, we can argue that it lies on the median MQ and NR. This means the centre of mass lies on the point of concurrence of the medians, i.e. on the centroid G of the triangle.
Hence, option **(3)** is the correct answer.

Q12) **Explanation:**

Hint: The moment of the force depends on the position of the force w.r.t the point about which the moment has to be calculated.

Step 1: Find the relative position of the force.
The position vector of the force is given by;
$\vec{r} = (2-2)\,\hat{i} + (0-(-2))\,\hat{j} + (-3-(-2))\,\hat{k}$
$\vec{r} = 2\hat{j} - \hat{k}$





Step 2: Find the moment of the force.
The moment of force (torque), is given by;
$\vec{\tau} = \vec{r} \times \vec{F}$
Subsitute the known values we get;
$\vec{\tau} = \left(2\hat{j} - \hat{k}\right) \times \left(4\hat{i} + 5\hat{j} - 6\hat{k}\right)$
$\vec{\tau} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 2 & -1 \\ 4 & 5 & -6 \end{vmatrix}$
$\vec{\tau} = (-12+5)\,\hat{i} - (0+4)\,\hat{j} + (0-8)\,\hat{k}$
$\vec{\tau} = -7\hat{i} - 4\hat{j} - 8\hat{k}$
Hence, option **(4)** is the correct answer.

Q13) **Explanation:**

Hint: $\tau = I\alpha$

Step 1: Find the angular acceleration of the wheel.
$\alpha = \dfrac{\tau}{I}$
$\tau = r \times F = 10 \times 0.1 = 1$ N-m
$\alpha = \dfrac{1}{0.1} = 10\,\mathrm{rad/s^2}$

Step 2: Find the angular velocity of the wheel.
$\omega = \omega_0 + \alpha t$
$= 0 + 10 \times 2$
$= 20\,\mathrm{rad/s}$
Hence, option (4) is the correct answer.

Q14) **Explanation:**

Hint: MOI of a spherical shell, $I = \frac{2}{3}MR^2$.

Explanation: The moment of inertia of a spherical shell about its diameter is given by the formulae: $I = \frac{2}{3}MR^2$.
Hence, option **(4)** is the correct answer.

Q15) **Explanation:**

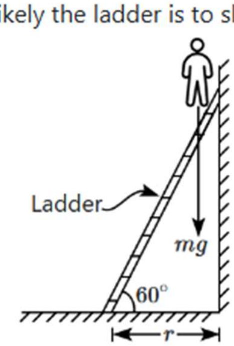
Hint: Use the parallel axis theorem.

Step: Find the moment of inertia of the system about the given axis.
The moment of inertia of a rod AB about an axis passing through A, perpendicular to the plane is given by;
$I_{AB} = \frac{mL^2}{3}$.
The moment of inertia of a rod CD about an axis passing through A, perpendicular to the plane, is given by;
$I_{CD} = \frac{mL^2}{12} + \frac{mL^2}{4}$
Therefore, the moment of inertia of the system about the given axis is given by;
$I = I_{AB} + I_{CD}$
$I = \frac{mL^2}{12} + \frac{mL^2}{4} + \frac{mL^2}{3}$
$\Rightarrow I = \frac{2mL^2}{3}$
Hence, option **(1)** is the correct answer.

Q16) **Explanation:**

Hint: The net torque increases as a person climbs on the ladder.

Explanation: When a person climbs a ladder, he exerts a force on the ladder that is directed upwards. This force tends to tip the ladder over, causing it to slip. The closer the person is to the top of the ladder, the greater the torque (twisting force) they exert on the ladder, and the more likely the ladder is to slip.



Hence, option(4) is the correct answer.

Q17) **Explanation:**

Hint: Radius of gyration, $k = \sqrt{\frac{I}{M}}$.

Step: Find the ratio of radii of gyration.

$$K_{\text{Disc}} = \sqrt{\frac{MR^2}{2M}} = \frac{R}{\sqrt{2}}$$

$$K_{\text{Ring}} = \sqrt{\frac{MR^2}{M}} = R$$

$$\frac{K_{\text{Disc}}}{K_{\text{Ring}}} = \frac{\frac{R}{\sqrt{2}}}{R} = 1 : \sqrt{2}$$

Hence, option (3) is the correct answer.

Q18) **Explanation:**

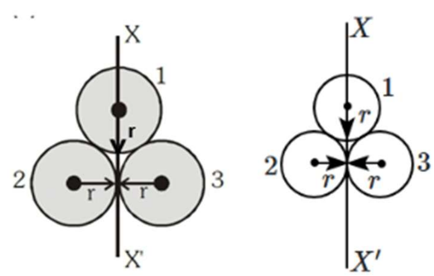
Hint: Use the parallel axis theorem.

Step 1: Find the moment of inertia of each shell about the given axis.

$$I_1 = \frac{2mr^2}{3} + mr^2 = \frac{5}{3}mr^2$$

$$I_2 = \frac{2mr^2}{3} + mr^2 = \frac{5}{3}mr^2$$

$$I_3 = \frac{2mr^2}{3}$$



Step 2: Find the net moment of inertia of the system.

The total moment of inertia of the system is given by;

$I_{\text{net}} = I_1 + I_2 + I_3$

$I_{\text{net}} = \frac{5}{3}mr^2 + \frac{5}{3}mr^2 + \frac{2mr^2}{3}$

$\Rightarrow I_{\text{net}} = 4mr^2$

Hence, option **(4)** is the correct answer.

Q19) **Explanation:**

Hint: $x_{com} = \frac{\int dm_i x_i}{\int dm_i}$

Step: Find the position of the centre-of-mass from A.
Let the position of the centre-of-mass from A is x.
The linear mass density of the rod is proportional to the distance from one end A i.e., $\lambda \propto x$
$\Rightarrow \lambda = kx$
The centre-of-mass of the system is given by;
$x_{com} = \frac{\int dm_i x_i}{\int dm_i} = \frac{\int_0^L (\lambda dx)x}{\int_0^L \lambda dx} \quad [dm = \lambda dx]$
$\Rightarrow x_{com} = \frac{\int_0^L (kxdx)x}{\int_0^L kxdx} = \frac{\int_0^L kx^2 dx}{\int_0^L kxdx}$
$\Rightarrow x_{com} = \frac{\frac{kx^3}{3}|_0^L}{\frac{kx^2}{2}|_0^L}$
$\Rightarrow x_{com} = \frac{2L}{3}$
Therefore, the position of the centre-of-mass from A is $\frac{2L}{3}$.
Hence, option **(3)** is the correct answer.

Q20) **Explanation:**

Hint: $F_{ext} = ma$

Explanation: The centre-of-mass of a system of particles moves as if all the mass of the system was concentrated at the centre-of-mass and all the external forces were applied at that point. From Newton's third law internal forces occur in equal and opposite pairs and the sum of internal forces and their contribution is zero. Only the external forces contribute to the motion of centre-of-mass.
Hence, option **(2)** is the correct answer.

Q21) **Explanation:**

Hint: MOI of a spherical shell, $I = \frac{2}{3}MR^2$.

Step: Find the relation between the MOI of given objects.
$I_0 = \frac{2}{3}MR^2$
$I_S = \frac{2}{5}MR^2$
$5I_S = 3I_0$
$\frac{I_S}{3} = \frac{I_0}{5}$
Hence, option (4) is the correct answer.

Q22) **Explanation:**

Hint: For a solid sphere, $I = \frac{2}{5}mR^2$; For a solid cylinder, $I = \frac{1}{2}mR^2$

Step 1: Find the radius of gyration of a solid sphere and a solid cylinder.
The radius of gyration is given by;
$\Rightarrow k = \sqrt{\frac{I}{m}}$
The radius of gyration of the solid sphere is given by;
$\Rightarrow k = \sqrt{\frac{2}{5}}R \quad \ldots(1)$
The radius of gyration of the solid cylinder is given by;
$\Rightarrow k' = \frac{R}{\sqrt{2}} \quad \ldots(2)$

Step 2: Find the ratio of the radius of gyration of a solid sphere and a solid cylinder.
From equations (1) and (2), we get:
$\Rightarrow \frac{k}{k'} = \frac{2}{\sqrt{5}}$
Hence, option **(1)** is the correct answer.

Q23) $I\omega_1 + I\omega_2 = 2I\omega$
$\omega = \frac{\omega_1+\omega_2}{2} \qquad \ldots(1)$

Also, we have energies of discs
$(K.E.)_i = \frac{1}{2}I(\omega_1)^2 + \frac{1}{2}I(\omega_2)^2$
$(K.E.)_f = \frac{1}{2}(2I)\omega^2$
from equation 1, we have
$(\mathrm{K.E})_f = \frac{1}{2}(2\mathrm{I})\left(\frac{\omega_1+\omega_2}{2}\right)^2$
$\text{loss in K.E} = (\mathrm{K.E})_i - (\mathrm{K.E})_f = \frac{1}{4}\mathrm{I}(\omega_1 - \omega_2)^2$

Q24) **Explanation:**

Hint: $\omega = \omega_0 + \alpha t$

Step 1: Find the angular acceleration of the disc.
$\omega = \omega_0 + \alpha t$
$\alpha = \frac{\omega-\omega_0}{t}$
$\alpha = \frac{100\times2\pi}{4} = 50\pi$ rad/s^2

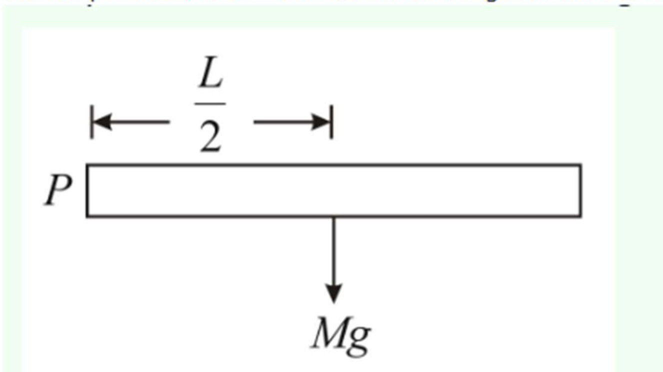
Step 2: Find the angle rotated during the given time interval.
$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$
$\theta = \frac{1}{2} \times 50\pi \times 16 \Rightarrow 400\pi$ rad
Hence, option (4) is the correct answer.

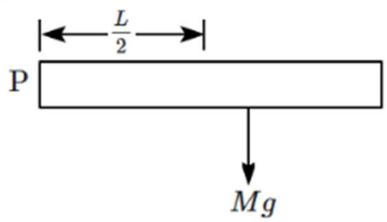
Q25) Hence, option (4) is the correct answer

Q26) **Explanation:**

Hint: Use the torque about point P to find the angular acceleration.

Step: Find the initial angular acceleration of the rod.
The torque on the rod = the moment of weight of the rod about P.





$\tau = Mg\frac{L}{2} \quad \ldots(1)$
The moment of inertia of the rod about the point P is given by;
$I = \frac{ML^3}{3} \quad \ldots(2)$
As, $\tau = I\alpha$
From the equations (1) and (2), we get;
$Mg\frac{L}{2} = \frac{ML^2}{3}\alpha$
$\Rightarrow \alpha = \frac{3g}{2L}$
Hence, option **(4)** is the correct answer.

Q27) **Explanation:**

Hint: $I = Mk^2$

Step 1: Find the radius of the gyration of a solid and hollow sphere.
$\frac{2}{5}MR^2 = Mk_S^2 \quad \ldots.(1)$
$\frac{2}{3}MR^2 = Mk_H^2 \quad \ldots.(2)$

Step 2: Find the ratio of the radius of gyration of a solid and hollow sphere.
from (1) & (2), we get:
$\frac{k_S}{k_H} = \sqrt{\frac{3}{5}}$
Note: The correct answer is $\sqrt{3} : \sqrt{5}$. which is not an option.

Q28) **Explanation:**

Hint: $\vec{v}_{com} = \frac{m_1\vec{v_1}+m_2\vec{v_2}}{m_1+m_2}$

Step: Find the velocity of the centre-of-mass.
The mass of each particle is m kg and particles moving towards each other with velocities $2v$ and v respectively.
The velocity of the centre-of-mass is given by; $\vec{v}_{com} = \frac{m_1\vec{v_1}+m_2\vec{v_2}}{m_1+m_2}$
$\Rightarrow \vec{v}_{com} = \frac{m(2v)-m(v)}{m+m} = \frac{mv}{2m} = \frac{v}{2}$
Therefore, the velocity of the centre-of-mass is $\frac{v}{2}$.
Hence, option **(3)** is the correct answer.

Q29) **Explanation:**

Hint: $x_{com} = \frac{\sum_{i=1}^{n} m_i x_i}{\sum_{i=1}^{n} m_i}$

Step: Find the centre of mass of the given system.
The center of mass of the square is located at its center, and the center of mass of the two point masses is at the midpoint of the line connecting them. Therefore, the center of mass of the given system lies on the line OY.
Hence, option (3) is the correct answer.

Q30) **Explanation:**

Hint: $\omega = \omega_0 + \alpha t$

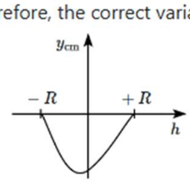
Step 1: Convert angular speed in rev/s.
Given, $\omega = 600$ rpm $= \frac{600}{60} = 10$ rps.

Step 2: Find the angular acceleration.
$\omega = \omega_0 + \alpha t$
$10 = 0 + \alpha(20)$
$\Rightarrow \alpha = \frac{1}{2}\ \mathrm{rev/s^2}$
Hence, option (4) is the correct answer.

Q31) **Explanation:**

Hint: $y_{\mathrm{cm}} = \frac{(M_{\mathrm{shell}} \cdot y_{\mathrm{shell}})+(M_{\mathrm{wax}} \cdot y_{\mathrm{wax}})}{M_{\mathrm{total}}}$

Step: Find the correct variation of the y_{cm} as a function of h.
The metallic shell is symmetric and has its centre-of-mass at $y = 0$.
Wax is added starting at the bottom of the sphere, with the centre-of-mass of the wax region below the centre of the sphere for small h. So, the overall centre-of-mass y_{cm} of the system will initially be negative because the wax dominates the lower region of the sphere.
As h increases,
For small h, most of the wax is concentrated at the bottom, pulling y_{cm} downward (negative).
For larger h: the wax starts to fill higher regions, and its centre-of-mass begins to rise, making y_{cm} less negative.
At $h = R$ (fully filled sphere): the system becomes symmetric, and $y_{\mathrm{cm}} = 0$.
Therefore, the correct variation of the y_{cm} as a function of h is shown in the figure below;



Hence, option **(2)** is the correct answer.

Q32) **Explanation:**

Hint: $\vec{\tau} = \vec{r} \times \vec{F}$

Explanation: When a person is high up on the ladder, a large torque is produced due to his weight about the point of contact between the ladder and the floor. Whereas when he starts climbing up, the torque is small. Due to this reason, the ladder is more apt to slip, when one is high up on it.
Therefore, Both (**A**) and (**R**) are True and (**R**) is the correct explanation of (**A**).
Hence, option (1) is the correct answer.

Q33) **Explanation:**

Hint: For balancing the bar, $\tau_{net} = 0$.

Step: Find the weight at the position P to balance the bar.
For balancing the bar, the net torque acting on the bar is zero i.e., $\tau_{net} = 0$.
By balancing the torque about the pivoted position we get;
$200 \times 1.5 + 1000 \times 0.5 - W \times 0.5 = 0$
$\Rightarrow W = 1600$ N
Therefore, the value of the weight at position P is 1600 N.
Hence, option **(4)** is the correct answer.

Q34) **Explanation:**

Hint: $\vec{v}_{cm} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1+m_2}$

Step: Find the velocity of the centre-of-mass before the collision.
$\vec{v}_{com} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1+m_2}$
$= \frac{1\times3+2\times0}{1+2}$
$\Rightarrow \vec{v}_{com} = 1$ m/s
Therefore, the velocity of the centre-of-mass before the collision is 1 m/s.
Hence, option (3) is the correct answer.

Q35) **Hint:** Center of mass is given by, $y_{cm} = \frac{m_1y_1 + m_2y_2}{m_1+m_2}$.

Step 1: Find the center of mass of the cylinder and the hemisphere.
Considering O as the reference point,
the center of mass of the cylinder lies in the middle i.e. at a distance $\frac{R}{2}$ below the point O.
The center of mass of the hemisphere lies at a distance $\frac{R}{2}$ above the point O.

Step 2: Find the center of mass of the system.
From point O, the distance of the center of mass of the system,
$y_{cm} = \frac{m\frac{R}{2} - m\frac{R}{2}}{m+m} = 0$
So the center of mass lies at the point O itself.

Q36) **Explanation:**

Hint: $v_{\text{CM}} = \frac{m_1v_1+m_2v_2}{m_1+m_2}$

Step 1: Find the horizontal component of the velocity of the centre of mass.
The horizontal component of the velocity of projectile motion is $v\cos\theta$. The velocity of the centre of mass is given as:

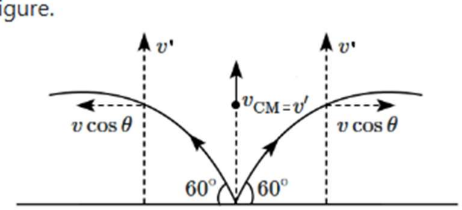
$$v_{\text{CM}} = \frac{m_1v_1+m_2v_2}{m_1+m_2}$$
$$= \frac{mv\cos\theta - mv\cos\theta}{m+m}$$
$$= 0$$

Therefore, the horizontal component of the velocity of the centre of mass is zero.

Step 2: Find the vertical component of the velocity of the centre of mass.
Let the vertical component of the velocity of the projectile at any particular time is v'. The velocity of centre of mass is given as:

$$v_{\text{CM}} = \frac{m_1v_1+m_2v_2}{m_1+m_2}$$
$$= \frac{mv'+mv'}{m+m}$$
$$= v'$$

Therefore, centre of mass of the projectiles moves along a vertical straight line as shown in the figure.



Hence, option (3) is the correct answer.

Q37) **Explanation:**

Hint: The position of the centre of mass is given by, $x_{cm} = \frac{m_1x_1+m_2x_2}{m_1+m_2}$

Step: Find the position of the centre of mass.
If the centre of mass lies at a distance of x from 2 kg mass.

$$x = \frac{m_1x_1+m_2x_2}{m_1+m_2}$$
$$x = \frac{2\times0+3\times1}{2+3} = 0.6\ m$$
$$\Rightarrow x = 60\ cm.$$

Hence, option (3) is the correct answer.

Q38) **Explanation:**

Hint: Angular momentum is given by, $L = I\omega$.

Step: Find the angular momentum of the rod.
The MOI of the rod about one of its ends = $\frac{1}{3}mL^2$
The angular momentum of the rod is given by;

$$L = I\omega$$
$$\Rightarrow L = \frac{1}{3}mL^2\omega$$

Hence, option **(2)** is the correct answer.

Q39) **Explanation:**

Hint: $\omega_f = \omega_i + \alpha t$

Step: Determine the time required to bring the given object to rest.

$\omega_f = \omega_i - \alpha t$

$\alpha = \frac{\omega}{t} \ldots (1) \quad (\omega_f = 0$ rad/s)

$\tau = I\alpha \ldots (2)$

From (1) and (2)

$$t = \frac{I\omega}{\tau}$$

Hence, option (2) is the correct answer.

Q40) Given,

A body of mass m is rotating about a fixed point

We Knows,

$$\vec{L} = m\left(\vec{r} \times \vec{v}\right)$$

So here, angular momentum is directed along a line perpendicular to the plane of rotation.

Q41) **Explanation:**

Hint: $I = \int r^2 dm$

Explanation: The moment of inertia is indeed dependent on how mass is distributed relative to the axis of rotation. In a hollow sphere, more mass is distributed farther from the center compared to a solid sphere, leading to a higher moment of inertia.

Therefore, Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.

Hence, option (1) is the correct answer.

Q42) **Explanation:**

Hint: Use the formula of torque.

Step 1: Find the angular displacement.

The angular displacement= 2π revolutions

$\theta = 2\pi \times 2\pi = 4\pi^2 \mathrm{rad}$

$\omega_0 = 3\mathrm{rpm} = 3 \times \frac{2\pi}{60} = \frac{\pi}{10} \mathrm{rad/s}$

$\omega_\mathrm{f} = 0$

Using the equation $\omega_\mathrm{f}^2 = \omega_\mathrm{i}^2 - 2\alpha\theta$

$0 = \frac{\pi}{10}^2 - 2\alpha 4\pi^2$

$\Rightarrow \alpha = \frac{1}{800} \mathrm{rad/s^2}$

Step 2: Find the torque required to stop a solid cylinder.

The torque of a solid cylinder is given by;

$\tau = I\alpha$

$\tau = \frac{MR^2}{2}\alpha$

Subsitute the known values we get;

$\tau = \frac{2 \times (4 \times 10^{-2})^2}{2} \times \frac{1}{800}$

$\Rightarrow \tau = 2 \times 10^{-6}\ \mathrm{N\,m}$

Hence, option **(2)** is the correct answer.

Q43) **Explanation:**

Hint: $\tau = \frac{dL}{dt}$

According to the principle of the conservation of angular momentum. When the resultant torque acting on a system is zero, the total angular momentum of a system remains constant.

Given, force $\overrightarrow{F} = \alpha\hat{i} + 3\hat{j} + 6\hat{k}$ is acting at a point $\overrightarrow{r} = 2\hat{i} - 6\hat{j} - 12\hat{k}$

As angular momentum about the origin is conserved i.e., τ= zero

$\Rightarrow \overrightarrow{r} \times \overrightarrow{F} = 0$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -6 & -12 \\ \alpha & 3 & 6 \end{vmatrix} = 0$$

$\Rightarrow$ (-36 + 36)$\hat{\mathbf{i}}$-(12+12α)$\hat{\mathbf{j}}$ + (6+6α)$\widehat{\mathbf{k}}$ = 0

$\Rightarrow$ 0$\hat{\mathbf{i}}$ -12 (1+α)$\hat{\mathbf{j}}$ + 6(1+α)$\widehat{\mathbf{k}}$ = 0

$\Rightarrow$6(1+α) = 0 $\Rightarrow$$\alpha$=-1

So, the value of a for angular momentum about the origin is conserved, α= -1

Hence, option (1) is the correct answer.

Q44) **Explanation:**

Hint: $X_{com} = \frac{m_1x_1+m_2x_2}{m_1+m_2}$

Step: Find the correct match.

For a uniform triangular lamina with height h, the centre-of-mass lies at $\frac{1}{3}$ of the height from the base.

For a solid cone, the centre-of-mass lies at $\frac{1}{4}$ of the height h from the base.

For a thin hemispherical bowl, the centre-of-mass lies at $\frac{1}{2}$ of the height h from the base.

For a solid hemisphere, the centre-of-mass lies at $\frac{3}{8}$ of the height h from the base.

Therefore, the correct match is A-II, B-III, C-I, D-IV.

Hence, option **(4)** is the correct answer.

Q45) **Explanation:**

Hint: $a = r\alpha$

Step: Find the acceleration of the rope.

Given: $F = 40$ N, $r = 0.5$ m, $\tau_{\text{friction}} = 10$ N-m, $I = 0.5$ kg-m^2

The net torque acting on the pully is given by;

$\tau_{net} = F \times r - \tau_{\text{friction}}$

$\Rightarrow \tau_{net} = 40 \times 0.5 - 10 = 10$ N-m

The net torque acting on the pully is given by;

$\tau_{net} = I\alpha$

$\Rightarrow \alpha = \frac{\tau_{net}}{I} = \frac{10}{0.5} = 20$ rad/s^2

As the rope does not slip on the pulley then, $a = \alpha r$

$\Rightarrow a = 20 \times 0.5 = 10$ m/s^2

Hence, option **(1)** is the correct answer.

Q46) **Explanation:**

Hint: Isotones have the same number of neutrons.

Statement(a) is not correct because isotones have the same number of neutrons .

Statement(b) is not correct as isobars differ in the number of neutrons because A is same but Z is different and neutrons = A - Z

Statement(c) is not correct because tritium $\left({}_{1}^{3}\mathrm{H}\right)$ is radioactive .

Q47) **Explanation:**

Hint: The formula of number of node is n-1

The formula of number of node is n-1.

For 5s, number of node is

5-1=4

For 5p, the number of node is

5-1= 4

For 5d, number of node is

5-1 = 4

Q48) Hint: Total nodes is 4.

Step 1:

Orbital has two radial as well as two angular nodes, which means the total node is 4.

The angular mode value is equal to l. Hence, l = 2. Thus, the orbital is d.

Step 2:

The formula of radial node is

Radial node = n – l – 1

2 = n – l – 1

l = 2

n – 2 – 1 = 2

n = 5

Q49) **Explanation:**

Hint: $KE_{\max} = h\nu - h\nu_0$

Q50) **Experience**:

Statement I:

"The energy of the He^+ ion in n=2 state is the same as the energy of H atom in n=1 state."

This statement is **true**.

- The energy levels of hydrogen-like ions can be described by the formula:

$$E_n = -\frac{Z^2 \cdot 13.6\text{eV}}{n^2}$$

$$For\ \text{He}^+(which\ has\ Z = 2) : E_2 = -\frac{2^2 \cdot 13.6\text{eV}}{2^2} = -13.6\text{eV}$$

$$For\ H(which\ has\ Z = 1) : E_1 = -\frac{1^2 \cdot 13.6\text{eV}}{12} = -13.6\text{eV}$$

Both energies are −13.6eV, confirming that the statement is correct.

Statement II:

"It is possible to determine simultaneously the exact position and exact momentum of an electron in H atom."

This statement is **false**.

- According to the Heisenberg Uncertainty Principle, it is impossible to know both the exact position and exact momentum of a particle at the same time. The more accurately you measure one, the less accurately you can measure the other.
- **Statement I**: True
- **Statement II**: False

Q51) **Explanation:**

Hint: Use uncertainty principle formula

Step 1:

The uncertainty in the speed is 2%. The speed of golf ball is 45 ms^{-1}. Calculate the uncertainty in speed as follows:

$$= 45 \times \frac{2}{100}$$

$$= 0.9\ ms^{-1}$$

Other given values are:

mass of golf ball = 40 g; = 40×10^{-3} kg

h = 6.626×10^{-34} J s

Step 2:

Calculate the value of Δx as follows:

$$\Delta x = \frac{h}{4\Pi m \Delta v}$$

$$= \frac{6.626 \times 10^{-34}\ \text{Js}}{4\times\ 3.14 \times 40\ \text{g} \times 10^{-3}\ \text{Kg}\ \text{g}^{-1}(0.9\ \text{m}\ \text{s}^{-1})}$$

= 1.46×10^{-33} m

Q52) Hint: $E_n = -13.6 \times \frac{Z^2}{n^2}$

$E_n = \frac{-13.6}{n^2}$ eV

For first Bohr orbit

n = 1, $E_1 = -13.6$ eV

n = 2, then will get $E = -3.4$ eV

Q53)

$$\text{P.E.} = \frac{1}{4\pi\varepsilon_0}\frac{(+Ze)(-e)}{r}$$
$$= \frac{1}{4\pi\varepsilon_0}\frac{(+3e)(-e)}{r} = -\frac{3e^2}{4\pi\varepsilon_0 r}$$

Q54) **Explanation:**

Hint: The number of emitted photoelectrons does not depend on the frequency of the incident light.

Explanation:

The statement that "the number of emitted photoelectrons increases with an increase in the frequency of incident light" is **not entirely accurate**.

Each material has a threshold frequency (ν_0), below which no photoelectrons are emitted, regardless of the intensity of the light. This frequency corresponds to the minimum energy required to liberate an electron from the material's surface.

As you increase the frequency of incident light (beyond its threshold), you provide more energy to each photon, which results in higher kinetic energy for emitted photoelectrons. This relationship is fundamental to understanding how light interacts with matter and forms a cornerstone of quantum mechanics and modern physics.

The number of emitted photoelectrons is independent of the frequency of incident light but the kinetic energy of emitted photoelectrons increases with an increase in the frequency of incident light.

Hence, **option 2** is the correct answer.

Q55) Bohr's radius for n^{th} orbit, $(r_n) = 0.529\,[\frac{n^2}{Z}]$ Å

where n = Energy level, Z = Atomic number

For 1^{st} Bohr orbit of hydrogen atom, n = 1 and Z = 1

Radius of H atom $r_H = 0.529$ Å

For He^+ ion: n = 2, Z = 2

$r_{He^+} = 0.529\,[\frac{2^2}{2}] = 2r_H$

For Li^{2+} ion: n = 2, Z = 3

$r_{Li^{2+}} = 0.529\,[\frac{2^2}{3}] = \frac{4}{3}r_H$

For Li^{2+} ion: n = 3, Z = 3

$r_{Li^{2+}} = 0.529\,[\frac{3^2}{3}] = 3r_H$

For Be^{3+} ion: n = 2, Z = 4

$r_{Be^{3+}} = 0.529\,[\frac{2^2}{4}] = r_H$

Be^{3+} (n = 2) has the same radius as that of the 1^{st} Bohr's orbit of hydrogen atom.

Q56) **Explanation:**

Hint: h orbitals require $l = 5$

Explanation:

The principal quantum number is the number that designates the electron shell where most of an atom's electrons are located. It also indicates the energy level occupied by the electron.

The azimuthal (or the orbital angular momentum) quantum number is defined as the determination of the shape of an orbital. Its value is equal to the total number of angular nodes in the orbital and is indicated by the symbol 'l.

For a given value of "n", the value of "l" can vary from 0 to n-1. The azimuthal quantum number for an h orbital is 5. So, the value of "n" (principal quantum number) will be 6.

Hence, **option 4** is the correct answer.

Q57) Hint: Order for filling of electron is 1s 2s 3s 3p 3d 4s 4p 4d 4f .....

Explanation:

Order of filling orbital is

$$1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f$$

33M $=1s^1 2s^1 2p^3 3s^1 3p^3 3d^5 4s^1 4p^3 4d^5 4f^7 5s^1 5p^2$

∴**Period=5**

Q58) Hint: The electrical discharge through the gases could be observed only at very low pressures and at very high voltages.

1. The electrical discharge through the gases could be observed only at very low pressures and at very high voltages. To achieve electric discharge in a gas, you need electrons to accelerate in the electric field and achieve kinetic energy high enough to ionize atoms. To this end, you need a high electric field and a long time between collisions, therefore, high voltage and low density.

2. In the absence of an electrical or magnetic field, these rays travel in straight lines.

3. The characteristics of cathode rays (electrons) do not depend upon the material of electrodes and the nature of the gas present in the cathode ray tube.

Hence, option second is the correct answer.

Q59) **Hint:** $r_n = \frac{a_0 \times n^2}{Z}$

Explanation:

STEP 1: The general formula to calculate the radius of an atom, $r_n = \frac{a_0 \times n^2}{Z}$

STEP 2: For the 3rd orbit, n = 3

Thus,

$r_3 = \frac{a_0 \times (3)^2}{3} = 3a_0$

$2\pi r = 3\lambda$

$2\pi(3a_0) = 3\lambda$

$\Rightarrow \lambda = 2\pi a_0$

$x = 2$

So, option 1 is the correct answer.

Q60) **Explanation:**

Hint: Number of protons = Mass number - atomic number

Assume the number of protons= x
Number of neutrons = $x + \frac{31.7x}{100} = \frac{131.7x}{100}$
Mass number = number of protons + number of neutrons
$81 = x + \frac{131.7x}{100}$
$x = 35$
Therefore, the symbol of the element is $_{35}X^{81}$

Q61) Hint: Compare the value of $\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

The energy needed for the transition of 3 to 8 is minimum among the other possibilities. So, energy frequency required for this transition is also minimum.

Q62) Apply (n+l) rule

5f>6p>5p>4d

Q63) **Explanation:**

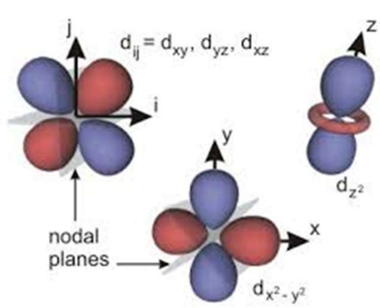
Hint: The d_{z^2} orbital has a doughnut-shaped region and two lobes along the z-axis.

Explanation:

The nodal plane is a plane that passes through a nucleus where the probability of finding an electron is almost zero. The number of nodal planes for any orbital is equal to the azimuthal quantum number is denoted by "l".

Most d orbitals have two nodal planes, but the d_{z^2} orbital is an exception. A d_{z^2} orbital has zero nodal planes; instead, it has two conical nodal surfaces due to its unique shape with a ring-like distribution in the xy plane and lobes along the z-axis.

This means that while there are areas of zero electron density in the d_{z^2} orbital, they are not flat planes like in other d orbitals.



Hence, **option 4** is the correct answer.

Q64) **Hint:** Orbital angular momentum = $\sqrt{l(l+1)}(h/2\pi)$

Explanation:

Step 1: Orbital angular momentum is given by $\sqrt{l(l+1)}(h/2\pi)$; "l" is the azimuthal quantum number.

It depends on the value of azimuthal quantum number.

Step 2: For 's' orbital "l" = 0

∴ Orbital angular momentum = 0

Thus, option 1 is the correct answer.

Q65) **Explanation:**

Hint: Number of orbitals in n^{th} shell= n^2
Explanation:
Number of orbitals in the n^{th} shell = n^2
For n = 4
Number of orbitals = 16

If each orbital is taken fully, then it will have 1 electron with m_s value of .
The number of electrons with m_s value of $\left(-\frac{1}{2}\right)$ is 16.

Q66) $E = E_0 + KE$
For minimum energy $E = E_0$
$$E = \left(\frac{hc}{\lambda}\right) = hv$$
$$= 6.6 \times 10^{-34} \times 1.4 \times 10^{+15}$$
$$= 9.24 \times 10^{-19}\ \mathrm{J}$$

Q67) Explanation

The graph represents the 2s orbital. It contains one node at $\psi = 0$.Total number of nodes=n-1

Q68) **Explanation:**

Hint: Use Heisenberg's uncertainty principle equation

The Heisenberg's uncertainty principle equation is as follows:

$$\Delta x . \Delta p \quad = \frac{h}{4\pi} \qquad or\ \Delta\ x.\text{m}\ \Delta\ v \quad = \frac{h}{2\pi}$$

$$\Delta v \quad = \frac{h}{4\pi \quad \Delta \text{x} \quad \text{m}}$$

The given values are as follows:

h = 6.626×10^{-34} J s

Δx = 0.1 Å

1 Å = 10^{-10} m

Δx = 0.1×10^{-10} m

m = 9.11×10^{-31} kg

Step 2:

$$\Delta v \quad = \quad \frac{6.626 \quad \times \quad 10^{-34} \quad Js}{4 \quad \times 3.14 \quad \times 0.1 \quad \times \quad 10^{-10} \quad \times 9.11 \quad \times 10^{-31} \quad kg}$$

= 0.579×10^{7} m s^{-1} (1 J = 1 kg m^2 s^{-2})

= 5.79×10^{6} m s^{-1}

Q69) **Hint:** Multiple transitions can be observed due to different energy levels.

Explanation:

Yes, a hydrogen atom produces several spectral lines even though it only has one electron in its orbit because the electron can transition between many different excited energy levels.

A hydrogen atom has one electron, but it has many energy levels, or shells, that the electron can occupy. When the electron jumps from one shell to another, it releases a photon. The energy difference between the shells causes different wavelengths to be released, which appear as spectral lines.

Both assertion and reason are correct, and reason is the correct explanation of assertion. Hence, **option 1** is the correct answer.

Q70) Hint: Bohr theory

The formula of the radius is as follows:

Where, n=number of the shell

Z= atomic number=2 (for He^+ atom)

For He^+ , Z value is 2

$\frac{a_0}{z} = \frac{52.9 \quad \mathrm{pm}}{2}$

$\mathrm{a}_0 = 26.5 \qquad \mathrm{pm}$

Q71) Hint: Use the formula that is, $E_n = -\frac{13.6}{n^2}$ eV.

The formula of energy of hydrogen atom is given below

$E_n = -\frac{13.6}{n^2}$ eV

For second excited state means, n = 3.

Calculate the energy in the second excited state as follows:

$$E_3 = -\frac{13.6}{9} = -1.51\mathrm{eV}$$

Q72) **HINT:** $E = \frac{Nhc}{\lambda}$

Explanation:

The law of photochemical equivalence states that a mole of reacting substance must have absorbed an equivalent quanta of light.

The energy is given as: $E = \frac{Nhc}{\lambda}$

So, $E = \frac{6.02 \times 10^{23} \times 6.62 \times 10^{-27} \times 3 \times 10^{10}}{\lambda}$

or, $E = \frac{1.196 \times 10^{8}}{\lambda}$

Thus, option 1 is the correct answer.

Q73) Hint: The valence shell electronic configuration of Z =114 is $7s^2\,7p^2$

Explanation:

The element with atomic number, Z = 114 is flerovium (Fl). It is a super heavy artificial chemical element. In the periodic table of the elements, it is a transactinide element in the p-block.

It is a member of the 7th period and is the heaviest known member of the carbon family.

Electronic configuration for Z =114 is $[Rn]_{86} 5f^{14}\,6d^{10}\,7s^2\,7p^2$

Thus, option 2 is the correct answer.

Q74) **Explanation:**

Hint:
Ni - 28 - $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^8$

Ni - $\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{\uparrow}\boxed{\uparrow}$ = 2 U.E.

Ni^{2+} - $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^8\ 4s^0\ 3d^8$

$\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{\uparrow}\boxed{\uparrow}$ = 2 U.E

Q75) **Explanation:**

Hint :

$Ni - 28 - 1s^2 2s^2 sp^6 3s^2 3p^6 4s^2 3d^8$

$Ni^{2+} - 28 - 1s^2 2s^2 sp^6 3s^2 3p^6 4s^2 3d^8$

$\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{\uparrow}\boxed{\uparrow}$ = 2 U.E.

$$\mu = \sqrt{n(n+2)} \quad BM$$
$$= \sqrt{2(2+2)} BM$$
$$= \sqrt{8} \quad BM$$

Q77) **Explanation:**
The velocity of the electron is inversely proportional to the radius of the orbit.
From Bohr's theory: $v \propto \frac{1}{n}$
The radius increases with the principal quantum number (n), so the velocity decreases.

Given that both statements are true, and the reason correctly explains why the velocity decreases in higher orbits, the correct answer is:
1. Both (A) and (R) are true and (R) is the correct explanation of (A).

Q78) $r \propto n^2$

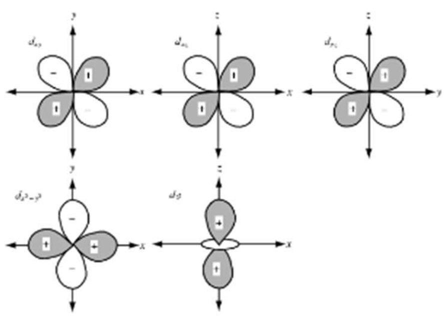
$$\frac{r_2}{r_3}=\left(\frac{2}{3}\right)^2=\frac{4}{9}$$

Q79) **Hint:** See the boundary surface diagrams of the five d orbital and find which d orbital has lobes in the axis.

Explanation:

d_{xy}, d_{yz} and d_{zx} orbitals have maximum electron density between the axis.

$\mathbf{d_{x^2-y^2}}$ and $\mathbf{d_{z^2}}$ have maximum electron density on the axis.



So, option 3 is the correct answer.

Q80) **Explanation:**

Hint: As the values indicate an orbital, a maximum of 2 electrons can be accommodated.

Explanation:

(i) The principal quantum number is a positive integer that indicates the electron shell, or energy level, that an electron occupies.

Here, the principal quantum number is 3.

(ii) The azimuthal quantum number **l** is a quantum number for an atomic orbital that determines its orbital angular momentum and describes aspects of the angular shape of the orbital. The value of l = 1. So, the subshell is **"p"**.

(iii) A magnetic quantum number is a quantum number used to distinguish the quantum states of an electron or other particle according to its angular momentum along a given axis in space. The orbital magnetic quantum number (m_l or m) distinguishes the orbitals available within a given subshell of an atom.

The magnetic quantum number m = -1.

This implies it is an orbital of the **3p sub-shell**. Any orbital can have a maximum of 2 electrons with a spin quantum number ±1/2. Thus, the maximum number of electrons with n=3,l=1, and m=1 is 2.

Hence, **option 3** is the correct answer.

Q81) To solve this, we can use the formula for the energy of an electron in the n^{th} energy level of a hydrogen-like ion:

$$E_n = -\frac{Z^2 R_H}{n^2}$$

Where:

- E_n is the energy of the electron in the n^{th} state,
- Z is the atomic number of the ion,
- R_H is the Rydberg constant,
- n is the principal quantum number

Given:

- The energy of the electron in the ground state (n=1) for the He^+ ion is x J.
- He^+ has Z=2.
- We are asked to find the energy for an electron in the n=2 state for the Be^{3+} ion, where Z=4.

Step 1: Energy of an electron in the ground state of He^+

For the ground state (n=1) of He^+:

$$E_1^{\mathrm{He}^+} = -\frac{(2)^2 R_H}{(1)^2} = -4R_H$$

This is given as x, so:

$$x = -4R_H$$

Step 2: Energy of the electron in the n=2 state for Be^{3+}

For Be^{3+} with Z=4 and n=2:

$$E_2^{\mathrm{Be}^{3+}} = -\frac{(4)^2 R_H}{(2)^2} = -\frac{16R_H}{4} = -4R_H$$

But we already know that $x = -4R_H$,so:

$$E_2^{\mathrm{Be}^{3+}} = -x$$

Thus, the correct answer is:

4. $-x$

Q82) **Hint:** Apply Heisenberg's Uncertainty principle

Explanation:

In the case of the hydrogen atom, the energies of the 2s- orbital and 2p- orbital are equal. This can be explained as follows:

Energy for hydrogen-like species can be given as,

$$\mathrm{E} = -13.6 \quad \frac{\mathrm{Z}^2}{\mathrm{n}^2}\mathrm{eV}$$

For 2s and 2p, n=2, so energy will be the same for 2s and 2p orbitals.

All other are correct statements. Thus, option 3 is the correct answer.

Q83) Hint: For any sub-shell (defined by 'l' value) 2l+1 values of m_l are possible

For l=5 total number of orbital is (2l+1)=11

Q84) Step 1:

The formula of energy is $E = \frac{hc}{\lambda}$. Energy is inversely proportional to the wavelength.

First convert each wavelength value in same unit.

λ(A) = 300 nm = 300x10^{-9} m=3x10^{-7} m
λ (B) = 300 μm= 300x10^{-6} m = 3x10^{-4} m
λ(C) = 3 nm = 3x10^{-9} m
λ (D) = 30 A° = 30x10^{-10} m = 3x10^{-9} m

Step 2:

Hence, the order of wavength is λ (B) >λ(A) >λ(C)=λ (D). The order of energy is (B) < (A) < (C)= (D).

Q85) **Explanation:**

Hint: It defines the three-dimensional shape of the orbital.

The azimuthal quantum number (l) is also called the subsidiary quantum number, denoted by l. It describes the shape of the orbital occupied by the electron.

Value of l	Subshell notation	Shape of orbital
0	s	spherical
1	p	Dumb-bell
2	d except dz^2	Double dumb-bell
3	f	Complex

Q86) **Explanation:**

Step 1: According to Bohr's model, the radius of an electron's orbit in a hydrogen-like atom is given by the formula: $r_n = n^2 r_1$

where: r_n is the radius of the nth orbit,

n is the principal quantum number,

r_1 is the radius of the first orbit.

Step 2: For the third orbit, the value of n = 3

So, $r_3 = 3^2 r_1 = 9r_1$

Hence, **option 4** is the correct answer.

Q87) **Explanation:**

Hint: Ionisation energy = $13.6 \times \frac{Z^2}{n^2}$

Explanation:

(i) The ionisation energy of hydrogen atom is $-13.6\ eV$.

For He^+, the atomic number is 2. The one valence electron is present in $1s$ subshell. But in excited state, $n = 2$

So, the energy of He^+ in excited state is: $-13.6 \times \frac{Z^2}{n^2}$

= $-13.6 \times \frac{4^2}{4^2} = -13.6\ eV$

(ii) The molecular orbital configuration of $H_2^+\ is\ \sigma_{1s}^1$ and for $He_2^+\ is\ \sigma_{1s}^2\ \sigma_{1s}^{*1}$

And the bond order is calculated as: $\frac{No.\ of\ bonding\ electrons - No.\ of\ antibonding\ electrons}{2}$

For H_2^+, the bond order is: $\frac{1-0}{2} = \frac{1}{2}$

Similarly, the bond order of He_2^+ is $\frac{2-1}{2} = \frac{1}{2}$

Both species will have the same bond order. Thus, **option 1** is the correct answer.

Q88) Hint: 2l + 1

The relation between n_m and l is as follows:

$n_m = 2l + 1$

$l = \frac{n_m - 1}{2}$

Thus, option second is the correct answer.

Q89) **Explanation:**

Hint: In Cu, a completely filled 3d orbital ($3d^{10}$) is more stable than a partially filled one ($3d^9$).

Explanation:

The atomic number and ground state electronic configuration for the given elements are as follows:

S.No.	Element	Atomic number	Electronic configuration
1	Cobalt	27	$[Ar]\ 3d^7\ 4s^2$
2	Nickel	28	$[Ar]\ 3d^8\ 4s^2$
3	Copper	29	$[Ar]\ 3d^{10}\ 4s^1$
4	Zinc	30	$[Ar]\ 3d^{10}\ 4s^2$

Copper has an exceptional electronic configuration due to the increased stability achieved by having a fully filled subshell. The Aufbau principle generally predicts that electrons fill the lowest available energy levels first.

However, for copper, which has 29 electrons, the expected configuration would be $[Ar]\ 3d^9\ 4s^2$. Instead, copper's actual configuration is $[Ar]\ 3d^{10}\ 4s^1$.

This exception occurs because a fully filled 3d subshell (with 10 electrons) is more stable than having 9 electrons in the 3d subshell and 2 in the 4s subshell. By moving one electron from the 4s orbital to the 3d orbital, copper achieves a completely filled 3d subshell, which is energetically more favorable due to lower exchange energy.

This stability arises from the fact that fully filled subshells have lower energy and provide additional stability to the atom. Hence, **option 3** is the correct answer.

Q90) Hint: Electronic transition between energy levels is the main reason for line emission spectra

The emission spectrum of a chemical element or chemical compound is the spectrum of frequencies of electromagnetic radiation emitted due to an atom or molecule making a transition from a high energy state to a lower energy state.

Line emission spectra are useful in the study of electronic structure because each element has a unique line emission spectrum.

Thus, second option is the correct answer

Mitoprep

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